Selective Residual Computation: A Causal Test of Refinement, Interference, and Gating

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Abstract

Behavioral scores underdetermine how a language model uses depth. Cycle A of this first experimental test of the Paper 0 dual-selection framework found that five-seed tiny causal decoders learned maximum and minimum goals but not modulo-10 sum. Every block was positively useful on the learned tasks; the sole replicated negative-utility cell occurred on the failed task and showed no later repair. Gates saturated near open and did not improve quality. Cycle B1 computed an exhaustive residual atlas; B2 separated self-attention (SA) and feed-forward (FF) writes; and B3 found no joint vector-plus-task repair in 12,150 trajectories. B4 trained 25 depth/capacity runs and found localized negative/near-zero utility without establishing competence-preserving slack. B5 repaired the task ecology using five value-selection goals: all five tasks and all ten pairs matched, 13 task-conditioned utility contrasts replicated, and minimum layer 8 had negative block and FF utility. B6 gates were causally consequential but matched ungated quality, tracked magnitude better than utility, and did not eliminate the negative layer. B7 then validated exact instrumentation parity on a real Qwen3-0.6B model and found positive mean SA/block utility at every layer on predictable sequences; no negative FF layer excluded zero. The exact gated release remained outside measured memory headroom, so no gated-Qwen result exists. Cycle B supports competence-matched task-conditioned computation and functional soft modulation, but not repair, strong interference, realized compute savings, a quality gain over baseline, or gated-Qwen transfer.

1 Scientific objective

Paper 0 defines the broader dual-selection framework. Paper 1 instantiates it in Transformer residual streams and asks whether residual blocks refine, complement, interfere, or can be modulated or skipped under task context. The scope here is experimental: loss, accuracy, exact match, and F1 remain primary behavioral outcomes, while internal quantities discriminate among mechanisms that could produce similar scores.

We therefore treat trained neural networks as experimental cognitive systems and ask:

1. What information becomes active?
2. What transformations become active?
3. How do representations evolve across token time and model depth?
4. When do successive residual updates refine, complement, or interfere with one another?
5. Can a distributed task/goal state predict which transformations will be useful?
6. Can task-dependent inhibition become real computation skipping?
7. Which strongest tiny-model findings survive a pretrained gated-attention comparison?

Backpropagation and token streams are held fixed. Biological analogy, local learning rules, and embodied tasks are outside this paper's evidential scope.

1.1 Questions and preregistered decision structure

The confirmatory sequence is deliberately asymmetric. E1 must establish a residual phenomenon before E3 is interpreted as a remedy, and E2 must establish task dependence before a goal-conditioned explanation is advanced. The primary hypotheses are:

1. **H1 (task-conditioned utility):** at least some blocks have utility $U_l(\tau)$ that changes across counterfactual goals with content held fixed;
2. **H2 (strong interference):** reliably negative utility, when present, co-occurs with later repair or compensation and replicates across seeds;
3. **H3 (learned modulation):** matched residual gates reduce activation of blocks independently classified by E1 as harmful or redundant more than random, static-route, or norm-based controls;
4. **H4 (dual selection):** in pretrained gated attention, gate values contain information about functional update consequence beyond attention concentration and update magnitude;
5. **H5 (complementarity):** useful high-gate updates need not have positive state–update cosine.

Thresholds for negligible utility, minimum effect size, bootstrap confidence level, and required seed replication are fixed in the experiment registry before final test inspection. Failure to reject a null is reported with uncertainty rather than converted into evidence for redundancy.

2 Mechanistic measurement framework

Let $x_{l,t} \in \mathbb{R}^d$ be the state entering a candidate transformation at layer l and token position t. Define the raw candidate update $\Delta_{l,t}$, computational-selection variable $g_{l,t}$ (or $g_{l,h,t}$), and effective update

$$\widetilde{\Delta}_{l,t} = g_{l,t} \odot \Delta_{l,t}, \qquad x_{l+1,t} = x_{l,t} + \widetilde{\Delta}_{l,t}.$$

For an ungated block, $g = 1$ by convention, but computational selection is claimed only through intervention. Pre-norm state, SDPA output, output-projection result, MLP output, and post-block state are named separately in artifacts and never compared as if they occupied the same location. The metric battery is deliberately broad, but each family must discriminate among mechanistic hypotheses rather than become an interpretability dashboard.

2.1 Memory activation

Reuse PRA-style measurements wherever the architecture exposes retrievable or materialized memory: active-to-available KV ratio; retrieved references, chunks, or gists; supporting-fact recall; routing precision/recall/rank; evidence coverage; materialization breadth; and task quality versus active-memory budget.

A central cross-line hypothesis is that memory utility need not be monotonic:

$$Q(k+1) < Q(k)$$

may occur when additional materialized memory introduces distraction or representational interference. Sparse activation may therefore be functionally superior rather than merely cheaper.

2.2 Attention concentration and dilution

Measure attention entropy, effective support

$$N_{\text{eff}} = \exp\left(-\sum_i \alpha_i \log \alpha_i\right),$$

top-k mass, evidence mass, distance profile, concentration indices, head similarity, and changes with prompt, context, or memory size. Attention describes interaction structure; it is not treated as a complete causal explanation.

Memory selection and computational selection are crossed into four empirical regimes: sharp-attention/low-gate, sharp-attention/high-gate, diffuse-attention/low-gate, and diffuse-attention/high-gate. Their frequencies and task consequences are reported by layer and head. H4 requires gate information not explained by attention statistics alone.

2.3 Residual refinement, complementarity, and interference

Use update norms and cosines, covariance, CKA/RSA, subspace angles, task-variable decodability, logit trajectories, and causal ablation.

Refinement. Successive updates improve a common evolving representation or response. Candidate evidence includes monotonic increase in correct-answer margin, increasing decodability of task-relevant variables, and positive causal utility of successive blocks.

Complementarity. A layer adds useful information without substantially undoing prior computation. Candidate signatures include low update alignment, increased representational rank or new decodable factors, and positive causal utility for both earlier and later transformations.

Interference. A transformation creates a state that later computation must repair, or its removal improves performance in a task-dependent way. For block F_l and task family τ define

$$U_l(\tau) = Q_\tau(\text{full}) - Q_\tau(\text{skip } l).$$

Then $U_l > 0$ is useful, $U_l \approx 0$ redundant, and $U_l < 0$ a candidate harmful transformation.

Strong evidence for interference requires three conditions:

1. statistically reliable negative task-conditioned utility $U_l(\tau) < 0$;
2. evidence that later computation reverses, compensates for, or repairs the affected trajectory;
3. replication across examples and seeds, not an isolated ablation artifact.

Geometry alone is never sufficient to label a transformation harmful.

Because block skipping can create off-manifold states, the confirmatory analysis includes graded attenuation and, where shape and scale permit, norm-matched replacement or counterfactual activation patching. Conclusions state the intervention used. Negative U_l from a single skip is an *interference candidate*, not strong interference.

2.4 Temporal and multiscale stability

Token position provides a temporal axis distinct from model depth. We characterize input embeddings, hidden states, heads, features, and learned subspaces over multiple windows W.

For each feature/subspace compute rolling mean, variance, skewness, kurtosis and their first/second differences; adjacent and separated chunk distribution distances; covariance and correlation-matrix drift; autocorrelation function (ACF) and ACF drift; lagged cross-correlation between features, heads, and layers; effective correlation length; CKA/RSA across temporal windows; principal-subspace and eigenspectrum drift; and layer×scale×token-position stability maps.

We explicitly search for *amplification–repair* events: a perturbation, task conflict, or nonstationarity grows at one layer and is reduced at a later layer. Multiple temporal spans are required because local windows can appear stationary while longer windows reveal topic, goal, or discourse transitions.

2.5 Head and layer organization

For any representation, attention, temporal, or causal similarity metric M, compare

$$\mathbb{E}[M(\text{heads within a layer})] \quad \text{with} \quad \mathbb{E}[M(\text{heads across layers})].$$

Add variance decomposition by layer/head/task, clustering, specialization, redundancy, causal ablation utility, and temporal stability. This tests whether functional organization is primarily layered, head-specific, task-specific, or distributed.

2.6 Conventional ML/DL and systems metrics

Retain loss/perplexity, accuracy, EM, precision, recall, F1, calibration, output entropy, logit margin, gradient norms and directions, weight norms, activation norms, update-to-weight ratios, training stability, FLOPs, realized latency, memory use, throughput, and active-block fraction. Internal metrics are scientifically useful only when connected to competence, robustness, or resource cost.

Soft gating is not counted as computation saving when the candidate transformation has already been evaluated. Active-FLOP and latency claims are restricted to hard routes that avoid execution; theoretical FLOPs and realized wall-clock speed are reported separately.

3 Datasets and task ecology

Dataset structure is part of the causal experiment. Paper 1 remains in NLP/token streams. Mazes, Sudoku, 2-D environments, and embodied agent tasks are deferred to a separate paper.

3.1 Synthetic natural-language counterfactual mixtures

The basic unit is a *counterfactual task family*. Construct content objects C_i that support multiple plausible intentions I_j:

$$(C_i, I_j) \to Y_{ij}.$$

The same content is reused for answering, explaining, extracting, classifying, critiquing, transforming, checking, planning, and action-selection-like textual responses.

Task identity must not collapse to a discrete `MODE` token. Use paraphrased and implicit instructions; goal evidence distributed across conversational context; audience, style, safety, resource, and output constraints; conflicting local affordances; controlled distance between goal-defining evidence and response-affording content; semantically continuous task families; distractors; and same-content/different-goal and same-goal/different-content counterfactuals.

Define a controllable *goal identifiability* variable measuring how strongly local lexical cues reveal the intended task. This allows direct tests of whether interference increases when the task must be inferred from distributed context. Ground-truth latent task factors are retained for probing and causal evaluation only.

3.2 PRA continuity datasets

Use WikiText-2, WikiText-103, HotpotQA, and QASPER where feasible. WikiText supports unconstrained token dynamics and temporal stability analysis; HotpotQA provides supporting-fact supervision for memory/evidence activation; QASPER provides long scientific-document QA. Reuse creates direct continuity with PRA routing, attention dilution, materialization, and long-context measurements.

4 Architectures and experimental priority

All architectures expose a common instrumentation record for embeddings, residual pre/post states, attention outputs/weights, MLP outputs, per-head outputs where feasible, logits, gradients, candidate/effective updates, and architecture-specific latent states. Model-specific hooks live behind an adapter. Native instrumented forwards must match untouched logits exactly when possible, or within a declared tolerance, before measurements are accepted.

4.1 Core discovery models

The mandatory first-cycle matrix is:

1. tiny decoder-only Transformer;
2. tiny Transformer + residual gates;
3. tiny Transformer + shared goal latent z;
4. tiny Transformer + z-conditioned residual gates.

Tiny models are the mechanism-discovery environment because they permit exhaustive block ablation, many seeds, dense metric capture, and controlled architectural intervention.

4.2 Selective pretrained transfer

The external comparison uses the released 1B baseline and headwise variants (model revision `aad415c45ec6`). Their full identifiers and revisions are pinned in the run config. The released configurations match on 28 layers, hidden width 2048, 16 query heads, 8 key/value heads, vocabulary size 152064, maximum position setting 32768, tokenizer family, and bfloat16 dtype. The headwise model adds one gate logit per query head to each attention query projection, so capacity is close but not parameter-identical. Shared release provenance does not prove identical data order or optimization history; comparisons are therefore described as closely matched pretrained contrasts, not a single-factor randomized experiment.

Verified gate location. Inspection of official implementation revision `f4c2a5f6ffd6` shows that the query projection packs query features and gate logits. For headwise gating the logits are reshaped to

$$g_l \in [0,1]^{B\times T\times H\times 1}$$

after a sigmoid. They multiply the per-head SDPA output after the head axis is moved to $[B,T,H,d_h]$ and before flattening and the output projection. Thus we distinguish the per-head ungated SDPA candidate $a_{l,h,t}$, its effective form $g_{l,h,t}a_{l,h,t}$, the post-projection attention write, and the residual state after addition. This sequence follows the release and the gated-attention study [5].

Pretrained gate interventions. Native gates are compared with forced-open, forced-closed, per-head mean, token-shuffled, and optional thresholded gates. Shuffling preserves marginal gate distributions. Native instrumentation is parity-tested separately from interventions. Candidate reconstruction from an effective output is rejected when a gate is too small for stable division; no clipped reconstruction is silently analyzed.

4.3 Conditional comparators

TRM, mHC, Mixture-of-Depths, LayerSkip, or other third-party architectures are conditional rather than mandatory for the first cycle. Choose based on the observed phenomenon: mHC if residual interference dominates; MoD/LayerSkip if conditional compute dominates; TRM if iterative refinement dominates.

5 Novel experimental mechanisms

The mechanisms are modular interventions, not assumed contributions.

5.1 Goal-conditioned residual gating

Replace

$$h_{l+1} = h_l + F_l(h_l)$$

with

$$h_{l+1} = h_l + g_l(z)F_l(h_l).$$

Start with scalar/block gates for causal clarity. Soft gates test modulation. Thresholded gates test actual skipping:

$$g_l(z) < \epsilon \Rightarrow F_l \text{ is not executed.}$$

5.2 Shared distributed goal latent

Introduce

$$z_{l+1} = z_l + G_l(z_l, h_l).$$

The goal latent is not supervised as a one-hot task identifier. It is learned through the task objective; probes are measurement devices. A key criterion is stronger than task classification: z should predict future causal utility,

$$z \to \widehat{U_l(\tau)}.$$

If z only encodes task semantics but cannot predict downstream computational requirements, it should not be interpreted as a useful control state.

5.3 Required goal-state controls

Compare z against current ordinary residual state h_l; pooled residual summaries over earlier layers/tokens; matched-dimensional random vectors with matched first/second-order statistics; shuffled-goal counterfactuals; a gate-only model; and a z-only model. These controls distinguish a genuine control state from generic additional capacity or dynamic sparsity.

5.4 Combined functional top-down modulation

Use

$$h_{l+1} = h_l + g_l(z_l)F_l(h_l), \qquad z_{l+1} = z_l + G_l(z_l, h_l).$$

"Top-down" is functional rather than literal backward layer connectivity: globally integrated task state regulates more local transformations.

5.5 Sufficient activation objective

Use

$$\mathcal{L} = \mathcal{L}_{task} + \lambda \sum_l c_l \phi(g_l)$$

and sweep λ. The strong hypothesis is not merely quality preservation under reduced compute. It is the existence of $C_1 \subset C_2$ such that

$$\mathrm{Cost}(C_1) < \mathrm{Cost}(C_2), \qquad Q(C_1) > Q(C_2).$$

This is the computation-side analogue of selective memory materialization.

6 Analysis and statistical design

The unit of analysis is declared for every result. Tiny-model claims use at least five independently trained seeds unless a resource exception is recorded. Example-level effects are paired within seed; seed summaries, rather than tokens treated as independent replicates, establish replication. Pretrained deterministic inference varies examples, task family, context condition, and prompt form. Closely matched but separately trained checkpoints are not analyzed as paired parameter realizations.

Primary estimates include sample count, mean, median where skewed, standard deviation, example-bootstrap confidence intervals, seed-level sign consistency, and effect size. Gate–metric relations include Pearson and Spearman correlations, binned curves, layer/head-stratified estimates, and pooled models with layer/head/task controls. A pooled correlation alone cannot support H4. Final test families are held out from architecture and threshold selection.

The minimum pretrained plot set is: gate distributions by layer and head; gate versus alignment, relative magnitude, downstream utility, and attention entropy; candidate versus effective-update geometry; intervention effects; and layerwise baseline-versus-gated dynamics. Plots preserve layer/head/token structure before aggregation.

6.1 Reproducibility and artifact contract

Every run stores config, seed, repository commit, environment, package versions, device/dtype, model and dataset revisions, tokenizer, context length, batch size, exact hook locations, gate semantics, and intervention mode. Derived observations retain run/model/dataset/task/example/token/layer/head identifiers and candidate/effective norms, cosines, attention statistics, quality, utility, and repair score. Large tables use Parquet; paper summaries use CSV. Raw activations are retained only when required to reproduce a derived result.

7 Cycle B—post-Cycle-A preregistration

Cycle A is a fixed empirical baseline, not a result to reinterpret. Its bounded conclusion is that a four-layer regime in which nearly every learned block had positive causal utility exerted little pressure for learned computational selection. The post-Cycle-A Cycle B work package was frozen before B1 result collection and tests whether this null is specific to shallow, capacity-constrained computation. Execution is ordered B1 residual atlas, B2 SA/FF decomposition, B3 stability, B4 depth/capacity sweep, B5 competence-matched task ecology, conditional B6 gating, B7 Qwen, and conditional B8/B9 Llama/Gemma replication. A later stage cannot repair invalid instrumentation from an earlier one.

The registered Cycle B hypotheses are:

1. **H-B1 (computational slack):** effective-depth fraction L_{eff}/L decreases once available depth exceeds necessary depth;

2. **H-B2 (sublayer specialization):** SA and FF writes have systematically different causal utility and residual signatures;
3. **H-B3 (task-conditioned computation):** competence-matched tasks have different block or sub-block utility profiles;
4. **H-B4 (stability signatures):** correct and incorrect trajectories differ in stability or replicated task-relevant repair;
5. **H-B5 (gating pressure):** gates become selective only where redundancy or interference is measurable;
6. **H-B6 (pretrained transfer):** at least some normalized tiny-model residual and SA/FF signatures replicate in Qwen;
7. **H-B7 (partial cross-family regularity):** some normalized signatures persist in Llama/Gemma while others remain architecture sensitive.

These hypotheses are post-Cycle-A but prospective for the Cycle B stages that test them. B1 is descriptive and validates the measurement stack; it cannot establish H-B1 or H-B2 because the stored Cycle A blocks combine SA and FF and all have depth four.

8 Staged experimental protocol

8.1 Core E1: baseline residual map

Train tiny baseline models across seeds; capture the common metric battery; exhaustively skip individual blocks and selected groups; compute $U_l(\tau)$; identify useful, redundant, and candidate harmful blocks; search for later repair; and measure task decodability, logit trajectories, attention dilution, head/layer similarity, and temporal stability. E1 establishes whether there is an architectural pathology worth fixing.

8.2 Core E2: counterfactual goal manipulation

Hold C fixed while changing distributed intent I. Measure

$$\Delta h_l = h_l(C, I_1) - h_l(C, I_2),$$

block utility, attention, logits, temporal statistics, and goal identifiability. Test whether task-conditioned utility changes systematically as goal evidence becomes less locally obvious.

8.3 Core E3: gated residual stream

Train matched baseline and gated tiny models. Compare task quality, harmful-block activation, repair signatures, gate entropy, task dependence, active FLOPs, realized latency where possible, and distractor robustness. Compare with random skipping, static task routes, magnitude/norm heuristics, and matched-capacity ungated controls.

8.4 Secondary E4: shared goal latent

Train z without gates. Test semantic organization, task-switch tracking, and especially prediction of future $U_l(\tau)$. Compare to residual/pooling/random controls.

8.5 Secondary E5: combined modulation

Combine z and gates. Test whether gates suppress blocks independently identified by E1 as harmful or redundant rather than simply learning arbitrary sparse routes.

8.6 Conditional E6: hard skipping and sufficient activation

Only after soft-gating structure is understood, convert strong inhibition into real execution skipping. Sweep ϵ and λ. Report Pareto fronts over quality, active FLOPs, realized latency, and measured interference.

8.7 Conditional E7: architecture transfer

First, run the released Qwen3-style baseline/headwise-gated pair on language modeling, retrieval or QA, multi-hop reasoning where supported, and the controlled counterfactual set. Test H4–H5 using native gates and gate interventions. Report conventional task metrics and the dual-selection regimes alongside internal measurements. Only then transfer the strongest tiny-model signature to whichever additional architecture matches the dominant finding.

8.8 Conditional E8: PRA memory–computation interaction

Where PRA is available, factorially vary memory materialization and active computation:

$$Q = Q(k_{\text{memory}}, k_{\text{compute}}).$$

Test whether excess memory increases residual interference and whether goal modulation changes optimal retrieval breadth.

9 Cycle A—Initial shallow-regime tests

9.1 E1: baseline residual map

We trained four-layer, width-64 causal residual decoders (150,758 parameters) on deterministic natural-language counterfactual families. Each content triple was paired with maximum, minimum, and modulo-10-sum goals expressed at three lexical-identifiability levels without a task-ID token. Content families were disjoint across 720 training, 180 validation, and 270 frozen test examples. Five independently initialized models used seeds 11, 22, 33, 44, and 55. Thresholds and the test split were fixed in `configs/e1_baseline.yaml` before test analysis.

Mean test accuracy was 0.612 (seed SD 0.039, range 0.570–0.663). The aggregate concealed a sharp task difference: maximum and minimum accuracy were 0.880 and 0.858, whereas modulo-10 sum accuracy was 0.098, effectively chance for the ten possible answer words. Accuracy did not vary monotonically with the lexical-identifiability manipulation (high 0.613, medium 0.631, low 0.591), so E1 provides no evidence that the nominal identifiability scale alone controlled difficulty.

Table 1: E1 task-conditioned block classification. Confidence intervals bootstrap seed means; a cell is candidate harmful only when its interval lies below -0.002 and at least four of five seed means cross that threshold.

Task	Useful	Uncertain	Candidate harmful	Strong interference
Maximum	4	0	0	No
Minimum	4	0	0	No
Modulo-10 sum	1	2	1	No

The sole candidate-harmful cell was the final block on modulo-10 sum: mean probability utility was -0.0073 with a 95% seed-bootstrap interval $[-0.0153, -0.0020]$, and four seed means were below the fixed -0.002 threshold. This does *not* support strong interference. The affected task was not learned, the fifth seed was near zero, and no example met the preregistered 25% downstream repair criterion. Moreover, a final block has no later block that could repair its write. We therefore retain this as a negative-utility observation on an underfit task and do not interpret it as a replicated interference mechanism.

Across the learned maximum/minimum tasks, every block was positively useful. Mean task–block utility ranged from 0.095 to 0.437 at the limits of the seed-level intervals. The first-cycle E1 result consequently favors distributed refinement over a pervasive harmful-block account. Per-example records, example-bootstrap intervals within seed, seed summaries, manifests, histories, and checkpoints are stored under `results/e1`. E2 will test the narrower surviving claim that utility changes with goal while content is fixed; it is not licensed to treat negative cosine or the isolated sum result as interference.

9.2 E2: same-content/different-goal counterfactuals

E2 reused the five frozen E1 checkpoints and paired all three goals within each of 90 test content families. The resulting 5,400 paired observations cover 90 families, three goal pairs, four layers, and five seeds. For each pair we measured the norm and cosine of the final-token residual difference, target-probability utility difference, target-probability difference, and attention-entropy difference. A goal effect required a seed-bootstrap interval beyond the preregistered absolute utility difference of 0.01 and at least four of five seed means beyond that threshold in the same direction.

Eight of twelve pair–layer contrasts met this rule: all four maximum–sum and all four minimum–sum contrasts. Their mean utility differences ranged from 0.138 to 0.262, and all five seed means had the same positive sign. None of the four maximum–minimum contrasts replicated. Their seed-bootstrap intervals crossed the 0.01 decision boundary, and their signs varied by layer and seed. Because maximum and minimum were both learned while sum remained at chance, the eight positive contrasts confound intended goal with task competence. We report that the literal all-task H1 trigger fired, but the competence-controlled evidence does not support task-conditioned block utility.

Residual states nevertheless differed substantially between same-content prompts: mean final-token $\|\Delta h_l\|$ increased from roughly 11 at the first block to 19–23 at the fourth, while paired-state cosines remained between 0.17 and 0.52. These descriptive differences cannot be assigned uniquely to inferred goal because the natural-language instructions themselves differ. The association between the nominal goal-identifiability score and block utility was negligible (Pearson $r = 0.012$; Spearman $\rho = 0.032$). The small Spearman p-value produced by treating repeated layer/seed observations independently is not used as evidence. Records and manifests are stored under `results/e2`.

E2 therefore narrows the surviving interpretation: the models represent counterfactual prompts differently, but this cycle does not show replicated goal-conditioned utility among tasks of comparable competence. E3 remains warranted as a mandatory architectural comparison, but not as a demonstrated remedy for strong interference.

9.3 E3: matched scalar residual gates

We trained five matched-capacity ungated controls and five scalar-gated decoders using the E1 data, architecture, training schedule, and seeds. Both variants contain exactly 151,018 parameters: the ungated control registers an unused parameter vector matching each gate's weight and bias. This isolates parameter count but does not equate optimization geometry, so the primary estimate remains the paired seed-level quality difference.

Native gated accuracy was 0.613 versus 0.612 for the matched ungated model. The paired difference was 0.0007 with a 95% seed-bootstrap interval $[-0.0074, 0.0089]$. Thus E3 provides neither a meaningful quality improvement nor degradation. Distractor-present accuracy was not worse than distractor-absent accuracy (0.628 versus 0.599), but this non-random prompt grouping cannot establish robustness.

Table 2: E3 test accuracy averaged across five seeds. All soft controls evaluate every block.

Condition	Accuracy	Condition	Accuracy
Matched ungated	.612	Native gate	.613
Forced open	.612	Token shuffled	.611
Task-static mean	.613	Random binary	.604
Candidate-norm rank	.612	Residual-norm rank	.613

The gates did not learn selective inhibition. Their mean was 0.977, median 0.981, and Bernoulli entropy 0.103 nats; only 1.69% fell below 0.9 and none fell below 0.75. Native-minus-forced-open accuracy was 0.0007 with interval $[-0.0022, 0.0044]$. The cell provisionally marked candidate harmful in E1 (final sum block) received a higher mean gate than all other cells (0.995 versus 0.975), contrary to H3. Gate–utility and gate–relative-magnitude Spearman correlations were -0.068 and 0.076. Their token-level p-values are not treated as inferential evidence because observations share examples, layers, and seeds; the effect sizes are small in either case.

Native gated, forced-open, and matched-ungated forwards took approximately 0.203, 0.194, and 0.198 milliseconds per example on the same GPU timing loop. These small differences are timing noise rather than conditional execution: every candidate block was evaluated, active-block fraction was 1, and soft gating saved no FLOPs. Shuffled, task-static, and norm-ranked controls reproduced native accuracy to within two tenths of a percentage point; random binary gates were slightly worse. The mechanistically relevant conclusion is therefore negative: this scalar-gated training run is nearly an ungated network, does not suppress the E1 negative cell, and does not support H3. Artifacts are stored under `results/e3`.

9.4 E4: shared goal latent and future utility

We trained five shared-latent models and five z-only controls. At every block, a 24-dimensional state was updated from its previous value and the final-token residual; no task label supervised z. The shared model added a projection of the final z to the output representation, whereas the z-only control predicted from that projection alone. Both variants contained 160,926 parameters.

The primary diagnostic fit fixed-α ridge probes on validation families and evaluated them on frozen test families. For each source layer, the probe predicted causal probability utility of every strictly later block. The 30 seed/source/target evaluations therefore avoid using the test split to fit probe weights. Ordinary current and token-pooled residual states, moment-matched random vectors, and independently shuffled goal states were evaluated under the same protocol.

Table 3: E4 prediction of future block utility on held-out content families.

Probe feature	Mean R^2	Median R^2	Mean MAE
Goal latent z_l	-.002	-.002	.148
Current residual	.031	.052	.142
Pooled residual	.163	.211	.131
Moment-matched random	-.167	-.157	.162
Shuffled z_l	-.153	-.151	.161

The goal latent beat random and shuffled controls, showing that it contains some structured signal, but it did not generalize as a useful future-utility predictor: mean $R^2 = -0.002$. Current residuals were modestly predictive and pooled residuals were substantially stronger. Relative to current and pooled states, z lost 0.033 and 0.165 mean R^2; it exceeded those controls by the fixed 0.01 margin in only 30% and 13% of evaluations. This fails the E4 shared-control-state criterion.

Behavioral interventions agree with the probe result. The shared model reached 0.593 mean accuracy, below the E1 baseline; shuffling its final goal contribution across examples left mean accuracy at 0.593, while zeroing it produced 0.596. The z-only model reached 0.547. These comparisons do not show that z is causally required for the learned behavior. The E1–E4 decision rule therefore selects the weak/null comparative-residual-dynamics path. E5 is retained as the prespecified combined-model falsification test; E6–E8 require a separate eligibility decision rather than automatic expansion. Artifacts are under `results/e4`.

9.5 E5: combined goal-conditioned modulation

Five combined models used the same shared z to condition a scalar gate at every block. They contained 161,026 parameters, so comparisons with the smaller baseline and single-mechanism variants are descriptive rather than capacity matched. The confirmatory analysis did not relabel blocks: it imported the E1 task–layer classifications before inspecting E5 gates.

Table 4: Mean test accuracy of the tiny-model variants across five seeds.

Variant	Accuracy	Variant	Accuracy
Matched ungated	.612	Gate only	.613
Goal + residual	.593	Goal only	.547
Goal-conditioned gate	.599		

The combined gates remained close to open (mean 0.957, median 0.956, entropy 0.173 nats). Mean gates were 0.982 for the E1 candidate-harmful cell, 0.941 for redundant/uncertain cells, and 0.957 for useful cells. Across seeds, useful-minus-harmful gate was -0.024 with a 95% interval $[-0.030, -0.019]$: the ordering is opposite to E1-aligned suppression. Because the sole harmful label is the final block of the unlearned sum task, this result cannot establish active preference for harmful computation; it does decisively fail the preregistered H3 direction.

Native combined accuracy was 0.599 (seed interval $[0.569, 0.624]$), compared with 0.595 forced open, 0.598 after shuffling goal state across the batch, and 0.596 after zeroing goal state. Gate–utility Spearman correlation was 0.013, while gate–relative-magnitude correlation was 0.150. The combined mechanism therefore neither improved quality nor aligned gates with independently measured utility. E5 closes the tiny-model cycle as a null/falsification result; it is not renamed generic dynamic sparsity because even meaningful sparsity was absent. Artifacts are under `results/e5`.

9.6 E6 eligibility: hard skipping not run

The preregistered E6 audit failed all enabling conditions. E1 did not support strong interference; E3's gated accuracy interval included zero; only 1.69% of gates fell below 0.9, against a fixed 10% minimum for a meaningful threshold sweep; and E5 did not show E1-aligned suppression. We therefore did not tune ϵ or λ on the test data and report no quality–compute Pareto curve. Thresholding nearly open gates after observing these nulls would be an unmotivated post-hoc experiment rather than evidence about sufficient activation. The decision and its source values are machine readable under `results/e6`.

9.7 E7 status: pretrained transfer not run

No pretrained behavioral or gate result is reported. The scientific eligibility audit was negative: E1 found no strong interference, E3 found no gated quality effect, E4's goal latent failed its ordinary-state controls, and E5 found no E1-aligned suppression. We nevertheless tested operational feasibility of the exact pinned headwise checkpoint rather than assuming it was unavailable.

The downloaded BF16 checkpoint at revision `aad415c45ec6` is 3,443,609,378 bytes and contains 311 tensors with 2,033,181,696 named parameters, including the tied output weight. The available device is a 4,294,836,224-byte GTX 950M. Ordinary loading was killed during CPU materialization under the available host-memory limit. A memory-mapped meta-model recovery then failed with CUDA out-of-memory during direct tensor placement in FP16; a native-BF16 runtime probe passed, but native-BF16 streamed placement also failed. The baseline checkpoint was therefore not downloaded.

Because the real model never completed a forward pass, the fixture-level adapter parity result is not promoted to real-checkpoint parity. Zero pretrained examples were evaluated, and there are no E7 task, gate, correlation, intervention, or baseline-comparison measurements. This is a resource and preregistered scientific stop, not a negative result about the released gated-attention model. Exact device, checkpoint, load-attempt, and decision metadata are stored under `results/e7`.

9.8 E8 eligibility: PRA factorial not run

The external PRA checkout contains the required memory-control implementation and 135 learned-routing checkpoints, so a memory-materialization axis exists in principle. At audit time it was an uncommitted `hybrid-pra` worktree at revision `f3aafd1500b5`; Paper 1 did not modify or silently pin that state. More importantly, E6 supplied no validated active-computation axis and E7 supplied no pretrained transfer measurements. Zero $Q(k_{\text{memory}}, k_{\text{compute}})$ cells were evaluated. Crossing real PRA budgets with post-hoc thresholds on nearly open tiny-model gates would not test the registered dual-selection interaction. E8 is therefore a preregistered stop, with the availability audit under `results/e8`.

10 Cycle B—Residual Dynamics Atlas and Capacity Expansion

10.1 B1: exhaustive residual-dynamics atlas

B1 reused, without retraining, all 30 stored Cycle A checkpoints: five each of baseline, matched baseline, scalar-gated, goal-plus-residual, goal-only, and goal-conditioned-gated variants. Each checkpoint was evaluated on the same 270 frozen test examples. At every non-padding token and four block locations we measured residual and candidate/effective-update norms, relative magnitude, state–update alignment, minimum novelty $1 - \cos^2(x, \Delta)$, dominance, displacement, direction drift, adjacent-write cosine and normalized cancellation. Attention entropy, top-k mass, effective support, sink score, evidence mass, and within-layer head similarity were computed per head. Full layer-pair residual/update cosine, cancellation, CKA, and RSA matrices used final non-padding tokens. Single-block target-log-probability utility was recomputed but Cycle A classifications were not relabeled.

The run computed 1,152,840 layer–token residual observations and 4,611,360 layer–head–token attention observations. All required derived metrics were finite. To avoid committing activation-scale data, artifacts retain 32,400 per-example final-token residual rows, 201,600 stratified attention summaries, full matrix cells, seed summaries, correct/incorrect partitions, configuration, model hashes, logs, and figures; no raw activation tensor is persisted.

The five baseline seeds show a consistent depth progression at the final non-padding token. Early candidate writes were anti-aligned with their entering state, while later writes were aligned:

Table 5: B1 baseline final-token geometry across all test tasks. Intervals bootstrap five seed means. Novelty is $1 - \cos^2(x, \Delta)$; direction is $\cos(x, x + \Delta)$.

Layer	Relative norm	State–write cosine	Novelty	State direction
0	.572 [.527,.609]	−.406 [−.449,−.342]	.817 [.782,.865]	.826 [.801,.850]
1	.642 [.594,.691]	−.105 [−.140,−.061]	.952 [.947,.959]	.823 [.809,.837]
2	.688 [.613,.783]	.115 [.080,.142]	.954 [.947,.961]	.846 [.822,.868]
3	.603 [.519,.683]	.329 [.305,.355]	.863 [.852,.876]	.902 [.887,.919]

This sign change is not interference evidence. Cycle A found positive causal utility for every block on the learned maximum/minimum tasks, including early blocks whose writes are anti-aligned. B1 therefore supplies a direct within-project counterexample to interpreting negative cosine as harmful computation.

Across the 30 stored checkpoints, off-diagonal residual-state cosine was .710 (descriptive checkpoint-bootstrap interval [.703,.718]), compared with .200 [.177,.220] for candidate updates. Residual-state CKA was likewise .652 [.641,.664], versus .288 [.264,.315] for writes; RSA gave .599 [.590,.609] versus .224 [.202,.249]. Adjacent-layer residual cosine was .838, whereas adjacent-write cosine was .227. The normalized pairwise cancellation score averaged .209 and did not by itself indicate repair or interference. These comparisons describe a persistent carrier state receiving more layer-specific writes, not a causal functional taxonomy. The 30-checkpoint intervals summarize the stored collection; some matched controls share seeds and training structure, so they are not population-level architecture intervals.

Gated variants reproduced Cycle A's near-open behavior in geometric form: the mean effective-to-candidate norm ratio was .977 [.971,.981] for scalar gates and .957 [.953,.959] for goal-conditioned gates, versus exactly one for ungated variants. Correct-minus-incorrect partitions produced some isolated task/layer differences, but direction and interval exclusion were not stable across tasks and depth; the underlearned sum task also reverses the competence balance. B1 therefore does not claim a general correctness signature. B3 will test preregistered stability/repair quantities with matched difficulty rather than selecting among these atlas cells post hoc.

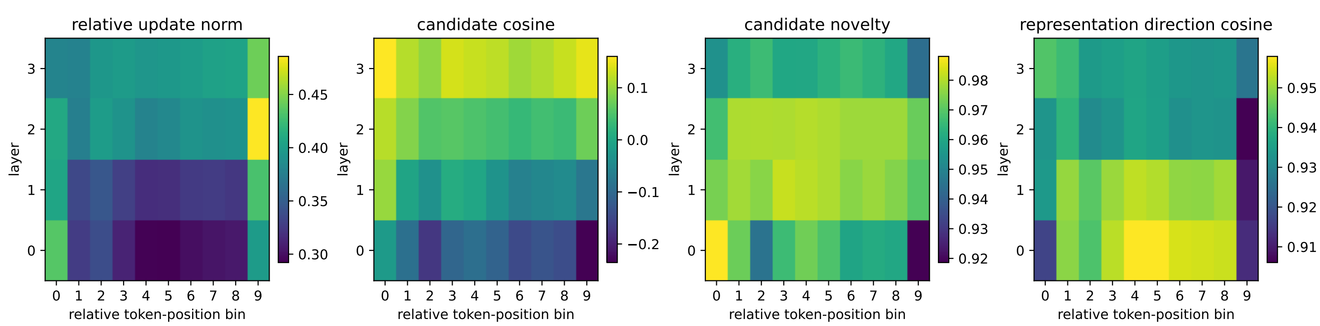

Figure 1: B1 layer–token atlas over correct examples from all stored Cycle A variants. Token position is binned relative to each non-padding sequence. Values are descriptive checkpoint/run summaries; causal utility remains a separate intervention quantity.

B1 passes its instrumentation gate: tensor locations are explicit, the full static battery is exercised, matrices are retained rather than collapsed to one scalar, and required outputs contain no non-finite values. The Cycle A nulls remain unchanged. B2 therefore proceeds to the mandatory separation of self-attention and feed-forward candidate writes, with native-forward and full-block parity tested before any sublayer result is accepted.

10.2 B2: self-attention versus feed-forward writes

B2 refactored the tiny pre-norm block without changing parameters or native computation. It records

$$x_l, \quad \Delta_l^{SA}, \quad x_l^{SA} = x_l + \Delta_l^{SA}, \quad \Delta_l^{FF}, \quad x_{l+1} = x_l^{SA} + \Delta_l^{FF}.$$

An architecture-neutral adapter exposes those locations. Native instrumented logits matched untouched logits exactly on every batch. Native and full-block-skip target probabilities also matched the persisted Cycle A records within 5.96×10^{-8}. Norm-matched random SA/FF replacements preserved per-token candidate norms within 1.91×10^{-6}. A zero replacement is exactly the corresponding skip path and is tested as an alias rather than counted as an independent intervention. Skip-SA recomputes the FF candidate from the intervened state x_l, whereas skip-FF retains the native SA write; this semantic difference is explicit in the artifact metadata.

Five frozen E1 baseline checkpoints produced 192,140 layer–token component observations and 5,400 paired example–layer causal observations. Utility uses target log probability. Across the eight competent maximum and minimum task–layer cells, both SA and FF utility intervals were positive. SA-minus-FF specialization replicated under the fixed 0.01 effect and four-of-five seed rule in seven cells; maximum layer 1 had a positive mean difference but failed seed sign replication. No cell met the intra-block repair criterion.

Table 6: B2 replicated SA–FF specialization across four layers per task. The modulo-10 task remained unlearned and is not used for competence-controlled task claims.

Task	SA > FF	FF > SA	Not replicated	Repair candidate
Maximum	3	0	1	0
Minimum	4	0	0	0
Modulo-10 sum	0	2	2	0

The causal scale of the distinction was substantial on learned tasks. Mean SA utility across layers was .609 for maximum and .420 for minimum, compared with FF utility .079 and .069. Averaged over all layer interventions, accuracy changed from .612 native to .491 under skip-SA and .603 under skip-FF; full-block skip gave .491. Norm-matched random replacement gave .476 for SA and .591 for FF. These pooled accuracies are descriptive because layer interventions are repeated within examples; the confirmatory result is the seed-paired task–layer utility analysis.

The failed modulo-10 task again supplies a useful negative control rather than positive theory evidence. SA utility was reliably negative at layers 0 and 2, while FF utility was small and positive; those two cells therefore registered FF > SA. However, their full-block utilities were also negative or near zero, the task remained at chance, mean SA–FF cosine was positive rather than compensatory, and no later task-relevant recovery criterion was met. They are not called intra-block repair or strong interference.

SA and FF also differed descriptively. Across correct final-token bins, relative SA write magnitude increased from .251 at layer 0 to .376 at layer 3, while FF decreased from .272 to .229. Both state–write cosines changed from negative early values to positive late values. Across-layer SA update cosine/CKA were .104/.196, compared with .364/.447 for FF writes, indicating that SA writes were more layer-specific under these metrics. Within-block SA–FF cosine was positive on average (.179 maximum, .153 minimum, .229 sum), and normalized cancellation remained near .20–.22; geometry therefore offers no hidden repair claim.

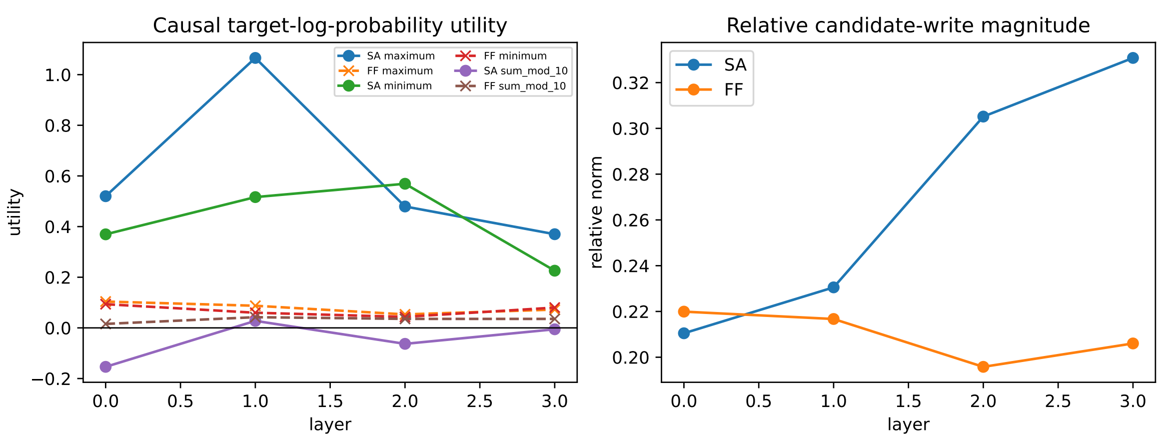

Figure 2: B2 causal SA/FF utility by task and relative candidate-write magnitude by depth. Skip-SA includes the downstream change in the recomputed FF write; utility is therefore a sublayer intervention effect, not the norm of the removed vector.

B2 supports H-B2 in this shallow regime: SA and FF have systematically different causal roles and residual signatures. It does not support H-B3: none of the eight maximum–minimum sublayer utility contrasts replicated under the competence-matched threshold. Nor does it reopen gating, because learned-task sublayers remained positively useful and no repair candidate was found. B3 therefore proceeds to the preregistered static/stability atlas without relabeling positive SA utility as selection pressure.

10.3 B3: token/depth stability and amplification–repair

B3 evaluated the same five frozen baselines with rolling windows 4, 8, and 16; first and second differences; ACF and correlation length; lagged cross-correlation; half-sequence mean, covariance, Wasserstein, and MMD drift; principal-subspace and eigenspectrum drift; CKA/RSA; and stable rank. The resulting 5,400 example–layer records keep correct and incorrect outcomes separate. For causal repair, every SA-skip, FF-skip, and full-block-skip perturbation at source layers 0–2 was tracked through later final-token states, producing 12,150 trajectories. A repair event required both later vector-divergence reduction and recovery of an intermediate output-head target-log-probability gap; norm shrinkage or logit-gap shrinkage alone was insufficient.

Depth changed several token-stability statistics. Mean residual-norm ACF(1) rose from $-.033$ at layer 0 to .211 at layer 3, while attention-entropy ACF(1) fell from .781 to .556. First-window residual stable rank fell from 7.35 to 5.32 (second-window 7.28 to 5.65). Half-window residual CKA fell from .756 to .640, while principal-subspace drift fell from .948 to .899. These are approximately stable or drifting coordinates under explicit operational definitions, not invariants or symmetries.

Twenty-three correct-minus-incorrect standardized cells passed the exploratory $|d| > .2$, four-of-five-seed rule. They were concentrated in learned tasks and late depth: maximum layers 2 and 3 contributed seven each, and 13 of 23 involved FF-update-norm statistics. For example, correct maximum trajectories had higher layer-2 FF-norm variance ($d = .748$, interval [.472,1.097]) and higher layer-3 FF-norm variance ($d = .537$, [.435,.620]); correct minimum trajectories had larger layer-3 residual norm ($d = .847$, [.509,1.159]). However, this battery contains many correlated comparisons, received no multiplicity correction, and was not difficulty-binned. B3 therefore records a coherent late-FF candidate signature for follow-up but does not treat 23 selected cells as confirmatory proof of H-B4.

The repair test is unambiguous. Intermediate target-log-probability gaps shrank after their peak in 63.8% of perturbation trajectories, but vector divergence never met the registered amplification–repair criterion; hence the joint repair rate was exactly zero. No task/source/intervention cell replicated above the 25% threshold. The frequent logit-gap reduction alone likely reflects downstream readout adjustment or saturation and cannot establish that later blocks restore the perturbed representation.

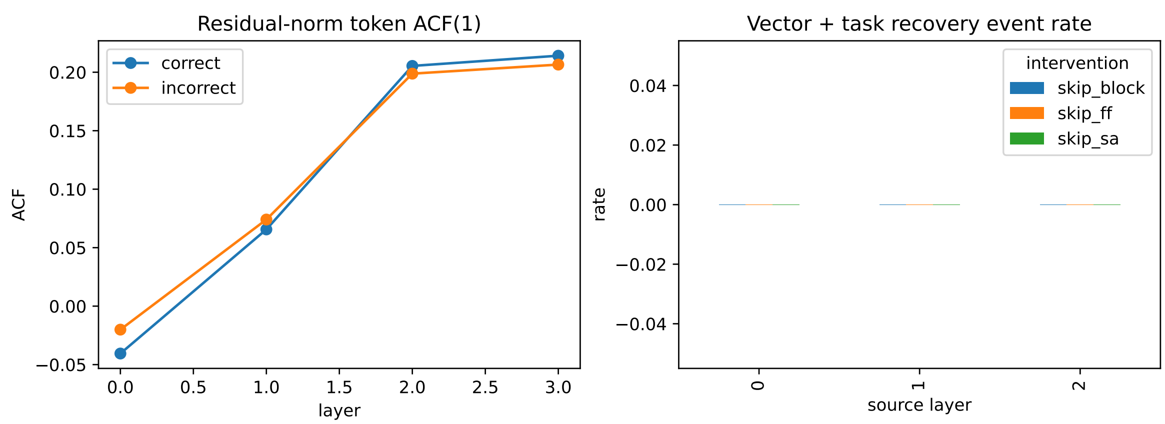

Figure 3: B3 residual-norm token ACF(1) by correctness and joint vector-plus-task recovery rates. No intervention trajectory met the joint repair criterion.

B3 leaves H-B4 unconfirmed: it supplies a replicating exploratory late-FF correctness signature but no repair signature and no matched-difficulty confirmatory contrast. Together B1–B3 still show useful distributed computation rather than redundancy or harmful writes on learned tasks, so B6 gating remains closed. B4 next tests whether adding depth creates the computational slack absent from the four-layer regime.

10.4 B4: controlled depth and capacity sweep

B4 trained width-64 baselines at depths 4, 8, 12, and 16, plus a 16-layer width-32 parameter-matched control, with five seeds per condition. The deterministic frozen split retained only maximum and minimum because Cycle A had not established competence on modulo-10 sum. The width-64 conditions contained 148,307 to 549,971 parameters; the matched control contained 143,955, close to the four-layer model's 148,307. Instrumented and native forwards had exact logit parity. Across 4,500 held-out predictions, B4 collected 50,400 example–layer causal records and 51,520 full pairwise-matrix entries.

Mean width-64 accuracy increased from .890 at depth 4 to .930, .942, and .948 at depths 8, 12, and 16. The narrow matched-parameter depth-16 control reached only .729, so depth did not compensate for the reduced width in this training regime. The registered competence rule required every main task/depth seed-bootstrap lower bound to reach .95; it failed, including at depth 4 and for depth-16 minimum. Consequently the decline in crude effective-block fraction from 1.00, to .938, .875, and .775 is reported as a candidate depth trend rather than competence-preserving computational slack. This effective count is based on independent single-layer ablations and is not a jointly minimal active subnetwork.

The causal decomposition did expose localized selection pressure. Minimum-task FF utility was reliably negative at depth-12 layer 4 (mean $-.00748$, 95% seed-bootstrap interval $[-.01414, -.00290]$) and depth-16 layer 13 (mean $-.00446$, $[-.00669, -.00235]$), with at least four of five seed means beyond the registered $-.002$ boundary. Minimum-task SA utility at depth-16 layer 14 was reliably near zero (mean $-.00049$, interval $[-.00114, .00011]$). These are sublayer intervention effects in models with approximately .94 minimum-task accuracy; they are not evidence of strong interference because no B4 trajectory met the joint vector-plus-task repair criterion. Pairwise geometry also changed with depth—off-diagonal update CKA rose from .329 at depth 4 to .524 at depth 16—but geometry is not assigned a causal label.

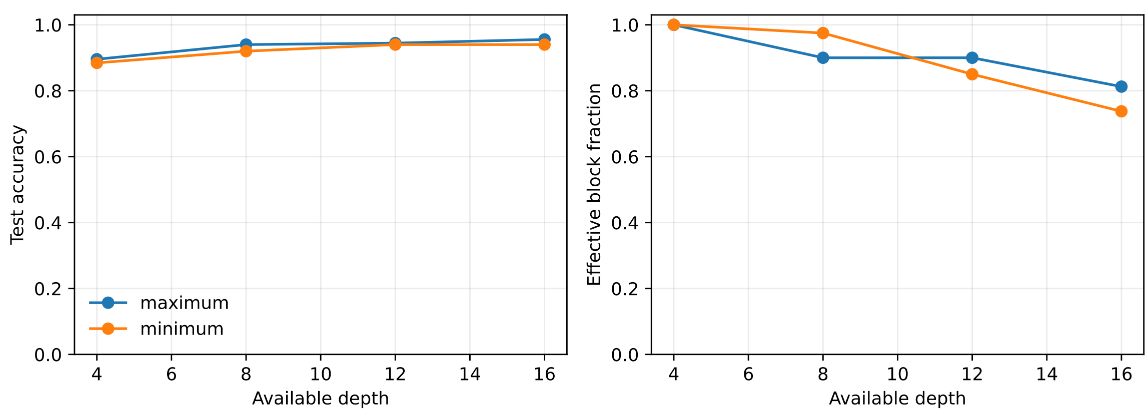

Figure 4: B4 held-out quality and independently ablated effective-block fraction. The falling fraction is not called competence-preserving slack because the registered all-condition competence gate failed.

B4 therefore supplies the first competent-task negative-sublayer and near-zero-sublayer candidates in this line, satisfying a Cycle B6 evidence trigger without supplying the repair required for an interference claim. Execution order is unchanged: B5 must first test whether these effects survive a larger, explicitly competence-matched goal ecology; B6 gating may then proceed against the evidence-selected deep conditions.

10.5 B5: expanded competence-matched task ecology

B5 began with validation-only difficulty control. A seven-task one-seed pilot excluded argmax-position and argmin-position at .567 and .583 validation accuracy. A five-task pilot retaining mixed high/medium/low instruction paraphrases still produced validation task means from .813 to .917 and no competence-matched pair. Confirmatory revision 3 therefore fixed direct high-identifiability wording for maximum, minimum, first, middle, and last. Pilot checkpoints remain audit artifacts, while their frozen-test measurements are excluded from all claims.

The confirmatory 16-layer width-64 baseline used five new seeds and exact native/capture parity. Held-out accuracies were 1.00 for first, middle, and last and .98 for maximum and minimum; every task cleared .90 in at least four seeds. All ten task pairs met the registered $|Q_i - Q_j| < .05$ rule in at least four seeds. B5 collected

24,000 example–layer causal records, 6,400 seed/pair/layer/token-bin goal-divergence cells, final-query attention redistribution, and validation-fit residual goal probes.

Thirteen competence-matched task-pair/layer/component contrasts passed the $|\Delta U| > .01$, four-of-five-seed, seed-bootstrap rule. They occupy seven pair–layer rows and comprise seven block and six SA contrasts; none is an FF contrast. For the original maximum–minimum pair at layer 4, maximum had lower utility by $-.126$ at block level (95% interval $[-.233, -.013]$) and $-.114$ for SA ($[-.214, -.014]$). Other replicated contrasts involved maximum–last at layer 1, maximum–middle at layer 3, and minimum versus first/middle/last at layers 1 or 4. Thus goal-conditioned causal roles emerge early in a competent ecology rather than from comparing learned and failed tasks.

B5 also replicated localized negative computation for minimum at layer 8:

$$U_8^{\text{block}} = -.0161, \qquad \text{CI}_{95} = [-.0277, -.0028],$$
$$U_8^{\text{FF}} = -.0105, \qquad \text{CI}_{95} = [-.0150, -.0057].$$

At least four seed means were below $-.002$. This is reliable harmful-under-ablation utility language, not strong interference: B5 does not add a replicated repair result. The validation-fit nearest-centroid linear goal probe rose from .265 at layer 0 to .739 at layer 5, then declined to .344 at layer 15 (chance .20). This supports goal information in the ordinary residual stream, not a separate causal controller; instruction wording and answer structure can both contribute to decodability.

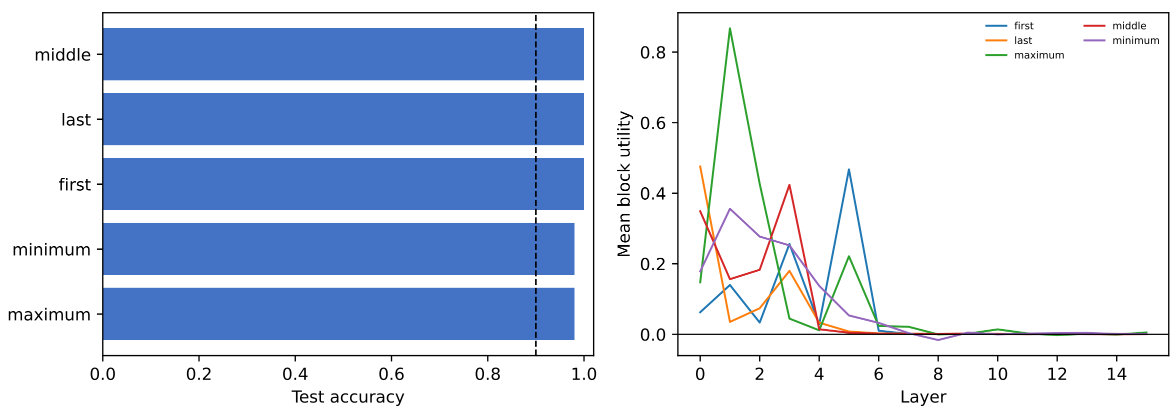

Figure 5: B5 confirmatory task competence and block-utility profiles in the 16-layer model. Only pairwise contrasts passing the independent competence rule enter task-conditioned claims.

B5 confirms H-B3 and keeps the B6 gate open on two independent triggers: competence-matched task-conditioned utility and replicated negative minimum-task block/FF utility. B6 should target the evidence-selected deep condition and compare whether learned gates track or mitigate these causal effects; it is not licensed to claim repair or general interference.

10.6 B6: evidence-triggered deep gating

B6 trained five whole-block input-gated and five SA/FF-specific input-gated 16-layer models on the B5 ecology. The component variant gates the SA write before constructing the FF input and then gates the FF write at its own pre-norm residual location. A shared-weight test verifies that forcing both component gates open reproduces the ungated baseline exactly; native capture parity was also exact in all ten trained runs. Each test example received native, forced-open, forced-closed, validation-mean, and within-batch shuffled controls plus independent block, SA, and FF skips, yielding 15,000 quality and 48,000 causal gate records.

Native mean accuracy was .991 for whole-block gates and .993 for SA/FF gates, compared with .992 for the B5 baseline. The paired seed intervals for gated-minus-baseline were $[-.006, .003]$ and $[.000, .002]$, far below the registered .02 meaningful-change threshold. Gating therefore did not improve held-out quality over an ungated model. It was nevertheless causally functional within the trained models. Forcing component gates open reduced accuracy by .024 (95% seed-bootstrap interval $[-.039, -.009]$); the whole-block decrease was .012 with interval $[-.033, .001]$. Replacing native values by validation means reduced accuracy by .010 and .011, while shuffling values reduced it by .028 and .025 for component and block gates. Fully closing either model reduced accuracy to .107.

Unlike Cycle A's near-open gates, deep whole-block gates averaged .797, with 12.3% below .5. Component SA and FF gates averaged .808 and .706; 12.9% of SA and 19.3% of FF gates were below .5. Gate–block-utility Spearman correlation averaged .259 for block gates and .200 for the component model. The corresponding gate–candidate-norm correlations were larger, .426 and .333; component SA/FF gate–utility correlations were .204 and .270. Thus gates carry task/example-dependent causal information, but update scale remains the stronger measured correlate.

The evidence-selected minimum layer 8 was not cleanly repaired or suppressed. In the component model, its mean FF gate was .653 while FF utility remained $-.0140$ and block utility $-.0236$; seed behavior was heterogeneous. The gates attenuate computation and their ordering matters to quality, but they do not selectively remove the B5 negative cell. Moreover, every candidate write is still evaluated, so these soft gates provide no realized FLOP or latency saving.

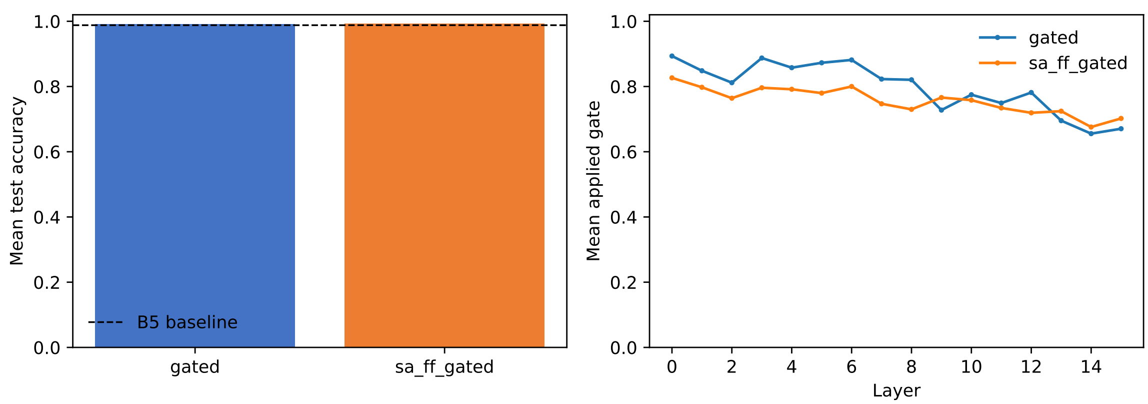

Figure 6: B6 native quality and learned gate profiles. The dashed line is the B5 ungated baseline; soft gate values do not represent skipped computation.

B6 supports functional, task-dependent amplitude modulation: component gates matter under forced-open, mean, and shuffled interventions. It does not support the stronger architectural claim of better quality, utility-aligned suppression, or active-compute reduction. B7 therefore proceeds as an external-validity probe using the same conservative tensor and intervention definitions.

10.7 B7: real small-Qwen probe and exact-gated resource stop

B7 first exercised the common adapter on the cached standard `Qwen/Qwen3-0.6B` revision `c1899de289a`. The model has 596.05M parameters and 28 layers. Hooks preserve the decoder-layer input, post-output-projection SA write, post-SA residual, MLP write, post-block residual, and per-head eager-attention weights. Untouched and instrumented forwards had exact logit equality. On 24 deterministic ascending, descending, alternating, trigram, weekly-cycle, and constant-sequence examples, B7 retained 10,192 layer–token residual rows, 163,072 layer–head–token attention rows, 5,488 full pairwise-matrix cells, and 2,016 paired SA/FF/block ablations. This is a fixed-model predictable-sequence probe, not a benchmark or a random sample of pretrained models.

Mean SA and full-block utility was positive at every layer. Twenty-five of 28 SA intervals and 24 of 28 block intervals excluded zero on the positive side. FF utility was positive on average in 25 layers; the negative means at layers 14, 17, and 18 were small ($-.0236$, $-.0261$, and $-.00894$ nats/token) and all bootstrap intervals crossed zero. Early layers dominated the loss effect, while later layers mostly supplied smaller positive changes. Across layers, relative FF-update magnitude correlated with FF utility at Spearman $\rho = .700$; the corresponding SA relation was .322. Mean attention entropy was essentially unrelated to SA utility ($\rho = -.001$). These associations are descriptive across 28 dependent layers and do not identify magnitude as the causal selection rule.

The matched released comparison did not fit this host. Static inspection re-verified revision `aad415c45ec6`: a query-derived sigmoid head gate multiplies SDPA head output before `o_proj`. Its cached weight file is 3.44GB and contains 2.033B named bfloat16 parameters (4.066GB of tensor storage). Before the standard model loaded, only 1.84GB host RAM and 3.58GB device memory were available; neither met the registered reserve-aware fit rule, and prior exact-checkpoint attempts had already failed CPU materialization and direct GPU placement. We therefore did not retry, download the release baseline weights, or collect gated metrics. Zero gated examples were evaluated and real gated-checkpoint parity remains unestablished.

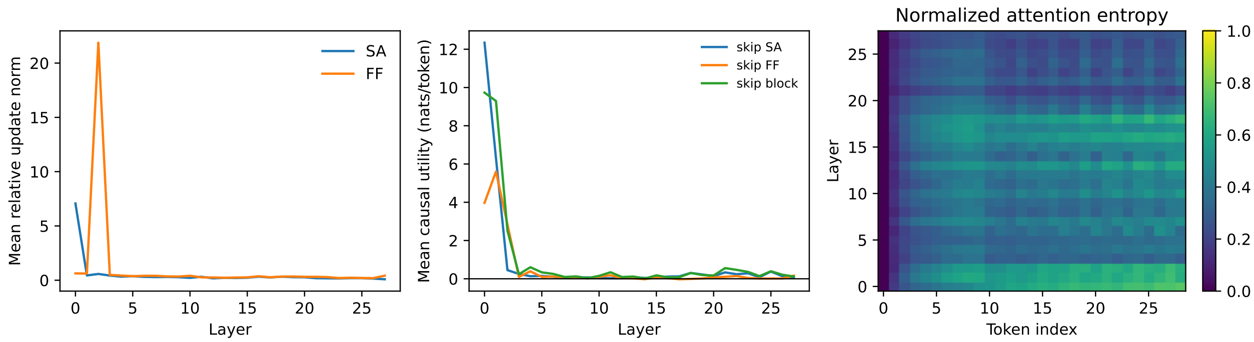

Figure 7: B7 standard Qwen3-0.6B residual geometry, paired causal utility, and layer–token attention entropy. This validates the real-model probe path but is not a gated-Qwen comparison.

B7 therefore gives bounded support to H-B6 only at the level of standard-Qwen measurement transfer: separate SA/FF signatures and causal asymmetry are observable with exact instrumentation parity. It does not test learned pretrained gates or establish gated transfer. Because the required Qwen comparison did not work end to end, the preregistered B8/B9 Llama/Gemma gate remains closed.

11 Primary claims, decision rules, and falsification

The paper starts with three nested claims.

Primary mechanistic claim. Residual blocks have task-dependent causal roles that can include refinement, complementarity, redundancy, and possibly harmful interference.

Secondary control claim. Distributed task/goal state predicts some variation in downstream block utility.

Stretch architectural claim. Goal-conditioned modulation exploits that structure to reduce interference or active computation and can improve quality or cost.

After E1–E4, choose the paper center by explicit decision rule:

- strong harmful/repaired block effects but weak gating $\Rightarrow$ interference characterization;
- strong goal-conditioned utility prediction $\Rightarrow$ distributed task-control state;
- strong modulation with quality gains $\Rightarrow$ goal-modulated residual computation;
- quality improves while compute falls $\Rightarrow$ sufficient/selective activation;
- weak/null effects $\Rightarrow$ comparative residual dynamics/falsification, with unsupported architecture claims removed.

Negative results are part of the design. If harmful blocks are rare, the interference hypothesis is weakened. If z does not beat ordinary residual summaries for predicting utility, the shared-control-state hypothesis is weakened. If gating only tracks update magnitude, it is interpreted as amplitude control. If attention statistics explain gates almost completely, the dual-selection claim is weakened. If forced-open gates barely change performance, native gate values are not described as causally selecting useful computation. If gating saves compute without changing interference, it is positioned as conditional computation rather than cognitive control.

12 Positioning and scope limits

This is a mechanism study, not a claim that Transformer gates instantiate biological inhibition. Representation similarity such as CKA [1], adaptive computation [2], dynamic depth such as Mixture-of-Depths [3], and LayerSkip [4] provide neighboring tools or comparators. The gated-attention release studies the performance, sparsity, stability, attention-sink, and long-context effects of post-SDPA gates [5]; the present contribution is to use its learned gate as an observable and intervenable variable in an architecture-independent residual-dynamics test. We do not attribute our cognitive interpretation to that work.

The present design cannot fully remove training-history confounds in pretrained comparisons, and block ablations may create off-distribution states. It addresses these limits through within-model interventions, closely matched release variants, graded controls, explicit tensor locations, and conservative causal language. Paper-series theory, embodied extensions, local learning, and detailed goal-latent architectures remain in Paper 0 and the roadmap.

13 Conclusion

Cycle A answers its bounded causal question with a constrained null. On tasks the tiny decoders learned, all four residual blocks contributed positive causal probability utility, while the only negative cell belonged to a chance-level task and lacked repair. Competence-matched counterfactual goals did not yield replicated utility changes. Scalar gates remained nearly open, neither gated variant improved quality or selected independently useful computation, and the explicit goal latent failed the stronger future-utility test against ordinary residual summaries.

Cycle B1–B7 expands what is known without rewriting that null. The residual carrier changes more smoothly across depth than candidate writes, exact SA/FF intervention shows causal asymmetry, and B3 finds no joint repair. B4 adds localized negative and near-zero sublayers but fails the strict slack criterion. B5 supplies the missing competence control: five tasks and all ten pairs match, 13 task-conditioned utility contrasts replicate, and minimum layer 8 has negative block/FF utility. B6 shows that deep learned gates are functionally consequential under forced-open and shuffled interventions, unlike Cycle A's saturated gates. However, native quality merely matches baseline, gate–utility alignment is weaker than gate–magnitude alignment, the negative cell persists, and no candidate computation is skipped. B7 validates exact real-model instrumentation and reproduces SA/FF causal asymmetry in standard Qwen3-0.6B, but the matched gated checkpoint remains unevaluated under an explicit resource stop. The combined result supports task-conditioned residual dynamics, soft amplitude modulation, and portable measurement semantics, not repair, strong interference, a quality–compute benefit, or gated/cross-family transfer. Conditional B8/B9 remain closed.

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